SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

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QUESTIONS: 10 **Total Marks: 50**

Annexure A
Application - Based Questions

Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to

define the state space, action space, and reward function to minimize waiting time and improve service quality. **(5 Marks)**

A ride-sharing company can use **Reinforcement Learning (RL)** to efficiently match drivers with passengers by learning the best assignment strategy from real-time data. The objective is to minimize passenger waiting time, reduce driver idle time, and improve overall service quality.

- **State Space:** Driver locations, passenger requests, traffic conditions, vehicle availability, estimated travel time, and demand level.
- **Action Space:** Assign a driver to a passenger, reject a request, or wait for a better match.
- **Reward Function:** Positive rewards for quick pickups, successful ride completion, high customer ratings, and reduced idle time. Negative rewards for long waiting times, ride cancellations, and inefficient driver allocation.

The RL agent continuously learns from ride history and traffic patterns, improving matching decisions over time, reducing operational costs, and increasing customer satisfaction

2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. **(5 Marks)**

An e-learning platform uses **Reinforcement Learning** to recommend personalized study materials based on each student's learning progress and interests.

- **Exploration:** The system recommends new topics or learning resources that the student has not previously accessed. This helps discover content that may improve learning.
- **Exploitation:** The system recommends materials similar to those that previously resulted in good learning outcomes and high engagement.

Balancing exploration and exploitation ensures students receive both familiar and new educational content. As the RL agent learns from quiz scores, completion rates, and feedback, recommendations become increasingly personalized and effective.

3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time.

(5 Marks)

A robotic vacuum cleaner uses **Reinforcement Learning** to clean rooms efficiently while avoiding obstacles and minimizing battery usage.

- **Policy:** The policy determines the best movement (move forward, turn left, turn right, recharge, etc.) based on the current environment.
- **Value Function:** The value function estimates the long-term reward of each state, helping the robot choose actions that maximize cleaning efficiency.

Over time, the robot learns which paths clean the maximum area with minimum energy consumption and fewer collisions, improving overall cleaning performance.

4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. **(5 Marks)**

A cryptocurrency trading system uses **Reinforcement Learning** to determine whether to buy, sell, or hold digital assets to maximize long-term profit.

Challenges in Reward Design:

- Cryptocurrency prices are highly volatile.
- Immediate profits may not represent long-term success.
- Balancing profit and investment risk is difficult.
- Delayed rewards make learning more complex.

Impact of Poor Reward Design:

- Encourages excessive or risky trading.
- May ignore long-term profitability.
- Can increase financial losses.
- Produces unstable or suboptimal trading strategies.

A well-designed reward function should consider profit, risk, transaction costs, and long-term investment returns.

5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.

A smart grid uses **Reinforcement Learning** to efficiently distribute electricity while balancing energy supply and demand.

- **State Space:** Electricity demand, renewable energy generation, battery storage level, grid load, and weather conditions.
- **Action Space:** Increase or decrease power distribution, store energy, purchase electricity from the main grid, or prioritize renewable sources.
- **Reward Mechanism:** Positive rewards for efficient energy utilization, reduced power loss, lower operational cost, and stable electricity supply. Negative rewards for overloads, outages, and energy wastage.

(5 Marks)

6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes. **(5 Marks)**

A healthcare monitoring system uses **Reinforcement Learning** to adjust medication dosages based on a patient's health condition.

- **States:** Patient's blood pressure, heart rate, blood sugar level, age, medical history, and previous medication response.

- **Actions:** Increase dosage, decrease dosage, maintain the current dosage, or recommend additional monitoring.
- **Rewards:** Positive rewards for improved patient health and stable vital signs. Negative rewards for side effects, incorrect dosage, or health deterioration.

The RL agent continuously learns from patient responses, enabling personalized treatment plans and improving long-term healthcare outcomes.

7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization.

(5 Marks)

A drone delivery system uses **Reinforcement Learning** to optimize navigation and delivery routes while avoiding obstacles and conserving battery power.

- **Exploration:** The drone tests alternative routes to discover faster or more energy-efficient paths.
- **Exploitation:** The drone uses previously successful routes that consistently produce good delivery performance.

A proper balance between exploration and exploitation enables the drone to adapt to changing weather, traffic restrictions, and obstacles, resulting in faster deliveries and reduced energy consumption.

8. **Design and develop** a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency.

Manufacturing robots use **Reinforcement Learning** to optimize assembly operations, improve production speed, and reduce errors.

- **Policy:** Determines the best sequence of assembly actions based on the current production stage.
- **Value Function:** Estimates the long-term reward of performing each action, helping the robot choose the most efficient assembly strategy.

As the robot gains more experience, it reduces assembly time, minimizes defects, increases production quality, and improves overall manufacturing efficiency.

(5 Marks)

9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance. **(5 Marks)**

Online advertising systems use **Reinforcement Learning** to maximize click-through rates (CTR) and advertising revenue by selecting the most relevant advertisements.

Reward Design Challenges:

- User clicks do not always lead to purchases.
- Delayed rewards make performance evaluation difficult.
- User interests continuously change.
- Multiple objectives such as CTR and revenue must be balanced.

Impact on Performance:
A poorly designed reward function may display irrelevant advertisements, reduce user engagement, lower advertising revenue, and negatively affect customer experience. A balanced reward function helps optimize both user satisfaction and business profit.

10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function. **(5 Marks)**

A smart irrigation system uses **Reinforcement Learning** to optimize water usage while maintaining healthy crop growth.

- **States:** Soil moisture level, weather conditions, temperature, humidity, crop growth stage, and water availability.
- **Actions:** Increase irrigation, reduce irrigation, stop watering, or delay watering.
- **Reward Function:** Positive rewards for healthy crop growth, water conservation, and higher agricultural yield. Negative rewards for water wastage, over-irrigation, and crop damage.

The RL agent continuously learns from environmental changes and sensor data, enabling efficient water management, reducing resource consumption, and improving agricultural productivity.

Code of Conduct:

I Saswanth (192425150) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Saswanth*

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately	Good understanding with minor gaps	Basic understanding; some	Poor understanding or misinterpretati

		identifies all requirements		requirements unclear	on of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand